

International Institute of Information Technology, Bangalore
CSE 102P Data Structures and Algorithms Lab- Sec A
Test1: 4 February 2026.

You are given an array `wealth` of n positive integers, where `wealth[i]` represents the wealth of person i . You are also given an integer k where $1 < k < n$.

A person is said to be **rich** if there are **not more than k people** who have wealth **greater than $k \times \text{wealth}[\text{person}]$** .

In other words, person i is rich if:

$$\text{count of people with wealth} > k \times \text{wealth}[i] \leq k$$

Your task is to find and return the number of rich people in the array.

Input Format

- The first line contains two integers n and k , separated by a space.
- The second line contains n space-separated positive integers representing the wealth array.

Output Format

A single integer representing the number of rich people.

Examples

Input	Output
6 2 10 3 5 20 1 4	4
5 1 5 10 2 8 1	2
4 2 2 2 2 2	4

Explanation of Examples

Example 1

For $k = 2$ and `wealth` = [10, 3, 5, 20, 1, 4]:

- Person with wealth 10: threshold = $2 \times 10 = 20$. People with wealth > 20 : none. Count = $0 \leq 2 \Rightarrow$ Rich
- Wealth 3: threshold = 6. People with wealth > 6 : [10, 20]. Count = $2 \leq 2 \Rightarrow$ Rich
- Wealth 5: threshold = 10. People with wealth > 10 : [20]. Count = $1 \leq 2 \Rightarrow$ Rich
- Wealth 20: threshold = 40. People with wealth > 40 : none. Count = $0 \leq 2 \Rightarrow$ Rich
- Wealth 1: threshold = 2. People with wealth > 2 : [3, 5, 10, 20]. Count = $4 > 2 \Rightarrow$ Not rich
- Wealth 4: threshold = 8. People with wealth > 8 : [10, 20]. Count = $2 \leq 2 \Rightarrow$ Rich

Total rich people = 4.

Example 2

For $k = 1$ and $\text{wealth} = [5, 10, 2, 8, 1]$:

- Wealth 5: threshold = 5. People with wealth > 5: [10, 8]. Count = 2 > 1
 $\Rightarrow$ Not rich
- Wealth 10: threshold = 10. People with wealth > 10: none. Count = 0 $\leq$ 1 $\Rightarrow$ Rich
- Wealth 2: threshold = 2. People with wealth > 2: [5, 10, 8]. Count = 3 > 1 $\Rightarrow$ Not rich
- Wealth 8: threshold = 8. People with wealth > 8: [10]. Count = 1 $\leq$ 1
 $\Rightarrow$ Rich
- Wealth 1: threshold = 1. People with wealth > 1: [5, 10, 2, 8]. Count = 4 > 1 $\Rightarrow$ Not rich

Total rich people = 2.

Example 3

For $k = 2$ and $\text{wealth} = [2, 2, 2, 2]$: All thresholds = $2 \times 2 = 4$. No one has wealth > 4, so count = 0 $\leq$ 2 for everyone. Total rich people = 4.

Constraints

- $2 \leq n \leq 2 \times 10^5$
- $1 < k < n$
- $1 \leq \text{wealth}[i] \leq 10^9$